

Smart Helmet — Sensor to ESP32 Wiring

ESP32 DOIT DevKit V1 · from *include/config.h* (imu-firmware) · I2C @ 100 kHz · schematic-only reference

Quick reference — all connections

Net / function	ESP32 pin	Modules
I2C SDA	GPIO22	ICM-20948, MAX30102, MAX30205
I2C SCL	GPIO21	ICM-20948, MAX30102, MAX30205
SPI MOSI	GPIO23	OLED DIN
SPI SCK	GPIO18	OLED CLK
OLED DC	GPIO2	OLED
OLED CS	GPIO5	OLED
OLED RST	GPIO4	OLED
MQ-135 AO	GPIO34	Air quality
MQ-136 AO	GPIO35	Air quality
MQ-4 AO	GPIO32	Methane
MQ-7 AO	GPIO33	Carbon monoxide
+3.3 V	3V3 pin	All sensors
Ground	GND pin	All sensors

Shared I2C bus (parallel wiring)

Sensor pin	Connect to	Notes
SDA	GPIO22	I2C data (all I2C devices)
SCL	GPIO21	I2C clock (all I2C devices)

Sensor pin	Connect to	Notes
VCC / VIN	3.3 V	ESP32 3V3 rail
GND	GND	Common ground

ICM-20948 (9-axis IMU) — I2C 0x68

Sensor pin	Connect to	Notes
SDA	GPIO22	Shared I2C bus
SCL	GPIO21	Shared I2C bus
VCC / VDD	3.3 V	
GND	GND	
AD0	GND	I2C address 0x68 (tie high for 0x69)
NCS / CS	3.3 V	I2C mode — must be high, not SPI
SDO / SA0	—	Leave default (I2C)

MAX30102 (heart rate / SpO₂) — I2C 0x57

Sensor pin	Connect to	Notes
SDA	GPIO22	Shared I2C bus
SCL	GPIO21	Shared I2C bus
VIN	3.3 V	
GND	GND	
INT	—	Not used in firmware

CJMCU-30205 / MAX30205 (temperature) — I2C 0x4F

Sensor pin	Connect to	Notes
SDA	GPIO22	Shared I2C bus
SCL	GPIO21	Shared I2C bus
VCC	3.3 V	
GND	GND	
A0 / A1 / A2	—	Straps set address 0x4F on your board

0.96" SPI OLED (SH1106)

Sensor pin	Connect to	Notes
DIN / MOSI	GPIO23	SPI data (VSPI)
CLK / SCK	GPIO18	SPI clock
DC	GPIO2	Data / command select
CS	GPIO5	Chip select
RST / RES	GPIO4	Reset
VCC	3.3 V	
GND	GND	

MQ-135 (air quality) — ADC1

Sensor pin	Connect to	Notes
AO (analog out)	GPIO34	ADC1 — air quality
VCC	3.3 V	
GND	GND	
DO	—	Not used in firmware

MQ-136 — ADC1

Sensor pin	Connect to	Notes
AO (analog out)	GPIO35	ADC1
VCC	3.3 V	
GND	GND	
DO	—	Not used in firmware

MQ-4 (methane) — ADC1

Sensor pin	Connect to	Notes
AO (analog out)	GPIO32	ADC1 — methane
VCC	3.3 V	
GND	GND	
DO	—	Not used in firmware

MQ-7 (CO) — ADC1

Sensor pin	Connect to	Notes
AO (analog out)	GPIO33	ADC1 — carbon monoxide
VCC	3.3 V	
GND	GND	
DO	—	Not used in firmware

Labelling tip: On the bench, label harness wires SDA→22, SCL→21, MOSI→23, SCK→18, MQ135→34, MQ136→35, MQ4→32, MQ7→33. Firmware may swap-retry SDA/SCL if init fails; working config is GPIO22=SDA, GPIO21=SCL.